Design Book Template

The Visualisation Design Book is where you document your data visualisation. The Design Book is the major piece of assessment for the Project that demonstrates that you have achieved the unit learning outcomes.

The following is a template to help you structure your Design Book.

See: **Project Design Book and Website (team) (Week 13)** **(https://swinburne.instructure.com/courses/78028/assignments/803538)** for assessment rubrics.

The following is a suggested outline for your Design Book:

Visualisation Design Book

Title Page

Includes:

- descriptive title (e.g., 'Data Visualisation Project' is not acceptable)
- team name and student names and IDs
- tutorial day and time
- year and semester
- word count

Table of Contents

1 Introduction

1.1 Background and Motivation

- Define the target audience (e.g., general public, analysts, policymakers, business users).
- Explain the importance of the visualisation. What problem does it address?
- Describe the key tasks users will perform using this visualisation.

1.2 Visualisation Purpose

- Define the specific questions users should be able to answer or what you are trying to achieve by making the visualisation
- List the benefits of the completed visualisation, explaining its impact on decision-making.

2 Data

2.1 Data Source and Governance

- Identify the original data source(s) and provide links where applicable.

- Include a brief data summary table (e.g., number of records, key attributes, update frequency)
- Discuss data governance, including:
  - Data collection process
  - Data quality assessment
  - Security, privacy, and ethical considerations
- Demonstrate how the data supports answering the questions from Section 1.2

2.2 Data Processing and Analysis

- List key attributes and their data types (categorical, ordinal, interval, ratio/quantitative)
- Describe the data cleaning process, including:
  - Handling missing values
  - Removing duplicates
  - Normalisation, transformation, and derived variables
  - Data filtering and joining datasets
- Include a screenshot of the KNIME workflow and append the .knfw file to the submission

2.3 Data Exploration

- Conduct exploratory data analysis using KNIME or other tools.
- Provide summary statistics and screenshots of exploratory data visualisations to understand patterns and outliers
- Discuss initial observations and any challenges encountered

3 Visualisation Design

This section should include:

3.1 Website Design

Create a wireframe of your webpage layout, showing:

- Navigation structure
- Placement of headings, text, and charts
- Provide a wireframe using tools like Figma, Miro, PowerPoint (or a hand-drawn sketch).

Create a storyboard showing how users will interact with your website and data visualisation to answer their question or achieve your purpose

3.2 Visualisation Design

- Explain the chart types chosen (e.g., bar, scatter, line, area, Sankey, parallel coordinates etc) and why they are appropriate.
- Discuss adherence to good design principles, including:
  - Graphical integrity (avoiding misleading charts)
  - Accessibility (colourblind and web-friendly palettes, font size)

    - Scalability (responsiveness across different screen sizes)
- Explain how graphical elements (colour, shape, size, annotations) are used to represent data differences effectively.
- Justify colour choices, labelling, and layout decisions.
- Describe how you will use annotations and tooltips enhance user understanding.
- Use sketches and/or screenshots to demonstrate design decisions.

3.3 Interaction Design

- Describe interactive features (e.g., zooming, filtering, tooltips, animations) and their role in improving user experience.
- Provide a table of interactions, explaining:
    - The interaction method (hover, click, drag, etc.)
    - The expected user behaviour and response

4 Iteration and Validation

4.1 Testing and refinements

- Describe the iterative process of testing and improving the visualisation
- Include feedback received (from peers, users, or instructors) and how changes were made in response
- Provide before-and-after comparisons using screenshots/sketches
- Discuss any adjustments made to design due to programming issues
- Discuss accessibility features of design/programming

4.2 Usability evlauation

- Conduct a usability evaluation

5 Conclusion and Future Improvements

Summarise key findings of project and lessons learnt.

Identification of possible future improvements.

References

References consulted (blogs, books, academic papers, discussion/help forums - for both design and programming)

Appendices

Gen AI Declaration

Usability evaluation test materials (if used)

Notes/data collected in usability evaluation (if used)